

Skill: GIT

File Name: 4.GIT-HOL

Master Branch Workspace Status Verification

```
MINGW64:/c/Users/hp/GitDemo

hp@Geethika MINGW64 ~
$ cd ~/GitDemo

hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout master
git status
Already on 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Feature Branch Content Modification

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout -b GitWork
echo "<note><heading>Reminder</heading><body>Don't forget this lab!</body></note>" > hello.xml
git add hello.xml
git commit -m "Add hello.xml in GitWork branch"
fatal: a branch named 'Gitwork' already exists
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[master 8a72e23] Add hello.xml in GitWork branch
1 file changed, 1 insertion(+), 1 deletion(-)

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Master Branch Divergent Asset Generation

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout master
echo "<note><heading>Alert</heading><body>This is a conflict test.</body></note>" > hello.xml
git add hello.xml
git commit -m "Add hello.xml with conflicting content in master"
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[master 6bc717e] Add hello.xml with conflicting content in master
1 file changed, 1 insertion(+), 1 deletion(-)

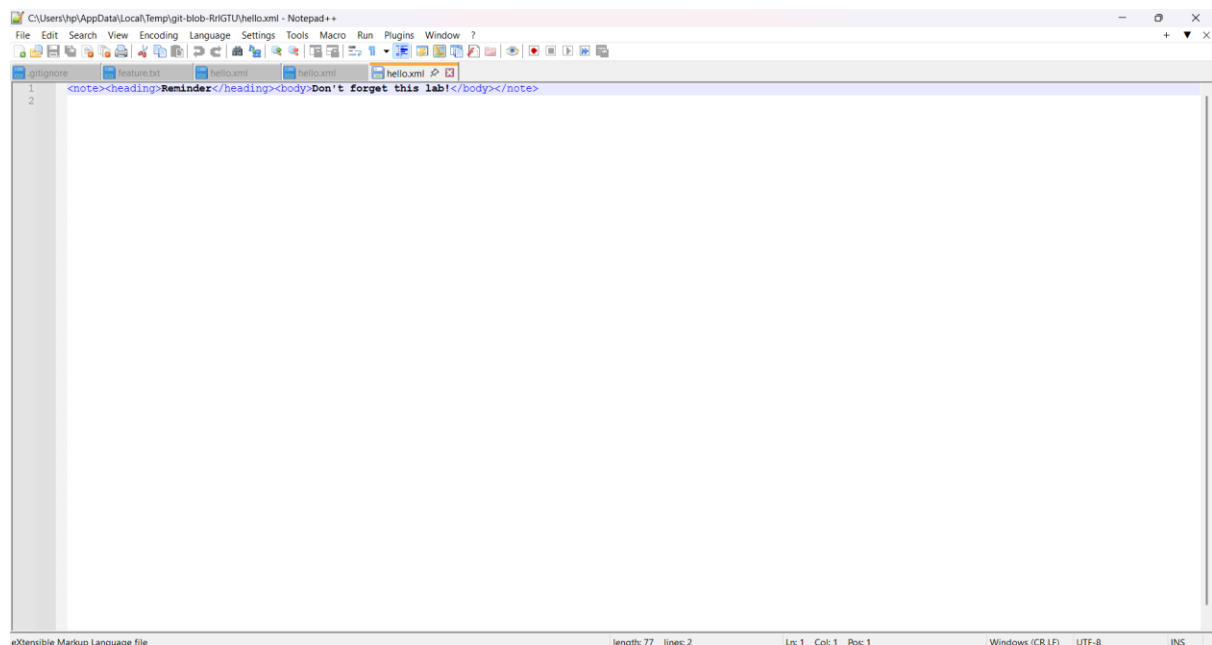
hp@Geethika MINGW64 ~/GitDemo (master)
$ |
```

Divergent Commit Graph Log Analysis

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
git difftool master GitWork
* 6bc717e (HEAD -> master) Add hello.xml with conflicting content in master
* 8a72e23 Add hello.xml in GitWork branch
* d6cf11c Add hello.xml with conflicting content in master
| * 6b02d4e (GitWork) Add hello.xml in GitWork branch
|/
* 4c1f60b (origin/master) Add feature.txt in GitNewBranch
* b70f066 Add .gitignore to exclude log files and folders
* a91c44c Initial commit for the GitDemo project

Viewing (1/1): 'hello.xml'
Launch 'p4merge' [Y/n]? y
```

Visual Inspection of File System Differences



Merge Conflict Trigger and Terminal Failure Catch

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

hp@Geethika MINGW64 ~/GitDemo (master|MERGING)
$ |
```

Manual Conflict Resolution and Staging Finalization

```
hp@Geethika MINGW64 ~/GitDemo (master|MERGING)
$ git add hello.xml
git commit -m "Resolved merge conflict in hello.xml"
[master 5f07b70] Resolved merge conflict in hello.xml

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Gitignore Calibration and Post-Merge Graph Analysis

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ echo "*.orig" >> .gitignore
git add .gitignore
git commit -m "Add diff tool backup files to gitignore"
git branch -d GitWork
git log --oneline --graph --decorate
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[master ac90a34] Add diff tool backup files to gitignore
 1 file changed, 1 insertion(+), 1 deletion(-)
Deleted branch GitWork (was 6b02d4e).
* ac90a34 (HEAD -> master) Add diff tool backup files to gitignore
* 5f07b70 Resolved merge conflict in hello.xml
| \
| * 6b02d4e Add hello.xml in GitWork branch
* | 6bc717e Add hello.xml with conflicting content in master
* | 8a72e23 Add hello.xml in GitWork branch
* | d6cf11c Add hello.xml with conflicting content in master
| /
* 4c1f60b (origin/master) Add feature.txt in GitNewBranch
* b70f066 Add .gitignore to exclude log files and folders
* a91c44c Initial commit for the GitDemo project
hp@Geethika MINGW64 ~/GitDemo (master)
$ |
```